17. Friday, July 26. Connectedness

Definition 1: Let X be a Topological Space.

1) A separation of X is a pair U, V of open sets such that

$$U, V \neq \emptyset, \quad X = U \cup V, \text{ and } U \cap V = \emptyset.$$

2) We say that X is connected if there does not exist a separation of X.

Lemma: Let X be a Topological Space. TFAE:

1) X is connected
2) for every open sets U, V with $X = U \cup V$ and $U \cap V = \emptyset$, we have either $U = \emptyset$ or $V = \emptyset$
3) The only subsets of X that are both open and closed (aka clopen) are $\emptyset$ and X.

Proof idea: (1) $\Leftrightarrow$ (2) is just a formal logical equivalence.
(2) $\Leftrightarrow$ (3) Follows from the fact that if U and V are subsets such that $X = U \cup V$ and $U \cap V = \emptyset$, then $V = X \setminus U$. $\square$

Examples:

1) Every set endowed with the Trivial Topology is connected.

2) Every discrete space with more than one element is not connected.
In fact $X = \{x\} \cup X \setminus \{x\}$ is a separation.

3) The subspace $Y = [-1,0) \cup (0,1]$ of $\mathbb{R}$ is not connected.

In fact, $U = [-1,0)$ and $V = (0,1]$ are open in Y, and form a separation of Y.

Lemma 2: Let X be a Top space and $Y \subseteq X$ a subspace. Then Y is connected if and only if for every open sets U, V of X such that $Y \subseteq U \cup V$ and $Y \cap U \cap V = \emptyset$, we have either $Y \subseteq U$ or $Y \subseteq V$.

Proof: HW!

Note: Proving that no separation exists can be challenging.

Example: Let $Y = [-1,1] \subseteq \mathbb{R}$. Then $U = [-1,0]$, $V = (0,1]$ satisfy $Y = U \cup V$ and $U \cap V = \emptyset$.
However, they don't form a separation, because U is not open in X.
We will show that Y is connected.

Proposition 3: If $\{A_i \mid i \in \Lambda\}$ is a collection of connected subspaces of a space X such that $\bigcap_{i \in \Lambda} A_i \neq \emptyset$, Then $\bigcup_{i \in \Lambda} A_i$ is connected.

Proof: Fix some $x \in \bigcap_{i \in \Lambda} A_i$ and denote $Y = \bigcup_{i \in \Lambda} A_i$.

Suppose $Y \subseteq U \cup V$ with $Y \cap U \cap V = \emptyset$. Since $x \in Y$, we have either $x \in U$ or $x \in V$ but not both. Assume wlog $x \in U$.

Claim: $Y \subseteq U$

Fix $i \in \Lambda$. Since A_i is connected, by lemma 3 we have either $A_i \subseteq U$ or $A_i \subseteq V$. But $x \in A_i$ and $U \cap V = \emptyset$, so we must have $A_i \subseteq U$. This holds for every $i \in \Lambda$, so also $Y = \bigcup_{i \in \Lambda} A_i \subseteq U$, as desired. ▷

Proposition 4: Assume A is a connected subspace of a space X.

Then every subspace B of X with $A \subseteq B \subseteq \bar{A}$ is also connected.

Proof: We use lemma 2.

Suppose U, V are open in X with $B \subseteq U \cup V$ and $B \cap U \cap V = \emptyset$.

We need to show that either $B \subseteq U$ or $B \subseteq V$.

Since A is connected and $A \subseteq U \cup V$ with $A \cap U \cap V \subseteq B \cap U \cap V = \emptyset$, by lemma 2

we know that either $A \subseteq U$ or $A \subseteq V$.

Assume wlog that $A \subseteq U$. Let us show that also $B \subseteq U$.

Let $x \in B$. To prove $x \in U$, since $B \subseteq U \cup V$ and $B \cap U \cap V = \emptyset$, it is enough to show

that x is <u>not</u> in V. Now, since V is open in X, we know that $X \setminus V$ is closed.

Further, this closed set contains A, because (using $A \subseteq U$) we have $\emptyset = A \cap U \cap V = A \cap V$.

Therefore $x \in B \subseteq \bar{A} = \bigcap_{\substack{C \text{ closed} \\ A \subseteq C}} C \subseteq X \setminus V$. Thus $x \in U$. It follows $B \subseteq U$, as desired. □

($X \setminus V$ is one of these)